Chapter 8: Application layer Services and Protocols

Application layer provides end user services using various protocols.

Need of Application layer design

- Think of different people/teams, working on the client and server programs.
  - Different programming languages.
  - Diverse hardware, operating systems.
- Allow for future extensions
  - Leave room for additional data, meta-data

Functions of Application Layer

- File Transfer: It allows user to access, retrieve and manage files
- Mail Services: Basis for email forwarding and storage
- Directory Services: Distributed database for global information about various objects and services

DNS (Domain Name System)

DNS is usually used to translate a host name into an IP address. It automatically converts the names we type in our Web browser address bar to the IP addresses of Web servers hosting those sites. Domain names comprise a hierarchy so that names are unique, yet easy to remember. DNS implements a distributed database to store this name and address information for all public hosts on the Internet. Most network operating systems support configuration of primary, secondary, and tertiary DNS servers, each of which can service initial requests from clients.

Example :

- Name used by human – www.example.com
- translated to the addresses 93.184.216.119 (IPv4)

Host name structure

- Each host name is made up of a sequence of *labels* separated by periods.
  - Each label can be up to 63 characters
  - The total name can be at most 255 characters.

Examples:

- whitehouse.gov
- Wikipedia.org
- Iamtheproudownerofthelongestlongestlongestdomainnameinthisworld.com

DNS Query Types

Recursive Query

A DNS client provides a hostname and DNS resolver must provide either a relevant resource or an error message if it can't be found

Iterative Query

A DNS client provides a hostname and DNS resolver returns the best answer. If DNS resolver has the relevant resource in cache, it returns them. If not refers to the root server or another Authoritative Name Server

Non-Recursive Query

In this case DNS resolver already knows the answer. It either directly returns the DNS record from cache or queries a DNS Name server which is authoritative for the record.

How DNS Works

When a user tries to access a web address like "example.com", their web browser performs a DNS query. The DNS server takes the hostname and resolve it into numeric IP address. The DNS resolver is responsible for checking if the hostname is available in local cache. If not contacts a series of DNS Name Servers, until it receives the IP of the service

Root name servers

It is contacted by local name server that can not resolve name. Root name server contacts authoritative name server if name mapping not known, gets mapping and returns mapping to local name server

TLD and Authoritative Servers

Top-level domain (TLD) servers:

Generic top level domains: responsible for .com, .org, .net, .edu, .gov, etc

Countries have a 2 letter top level domain: Example., uk, fr, np, ca, jp

Authoritative DNS servers

It is organization's DNS servers, providing authoritative hostname to IP mappings for organization's servers (e.g., Web and mail). It_can be maintained by organization or service provider

Local Name Server

It does not strictly belong to hierarchy. Each ISP (residential ISP, company, university) has one.
It is also called "default name server". When a host makes a DNS query, query is sent to its local DNS server. It acts as a proxy, forwards query into hierarchy.

HTTP and HTTPS

HTTP stands for Hyper Text Transfer Protocol . HTTP is an application-level protocol for distributed, collaborative, hypermedia information systems. This is the foundation for data communication for the World Wide Web (ie. internet) since 1990 . It allows the transmitting and receiving of information across the Internet

The S stands for "Secure" in HTTPS i.e., Secure hypertext transfer protocol. HTTPS is http using a Secure Socket Layer (SSL). Secure HTTP (HTTPS) is one of the popular protocols to transfer sensitive data over the Internet. A secure socket layer is an encryption protocol invoked on a Web server that uses HTTPS. Most implementations of the HTTPS protocol involve online purchasing or the exchange of private information. HTTPS is only slightly slower than HTTP

HTTP Connection

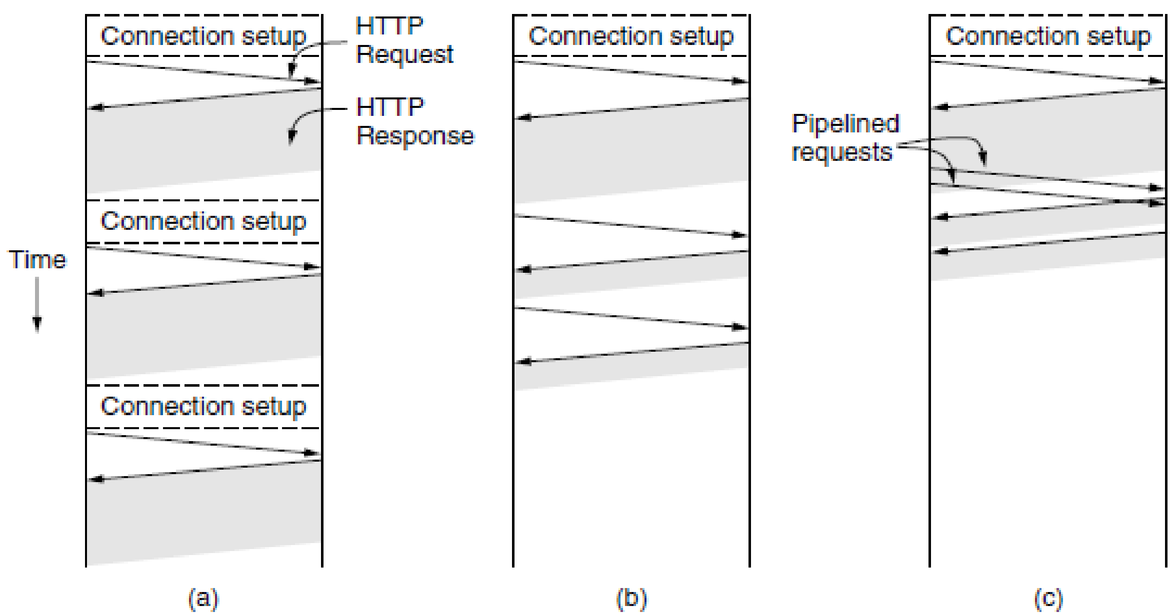


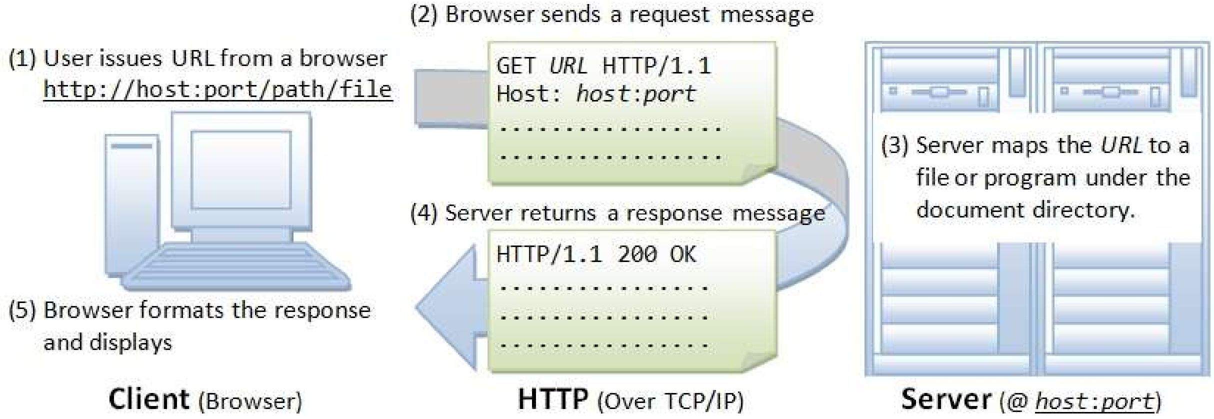
Figure: (a) multiple connections and sequential requests. (b)A persistent connection and sequential requests (c) A persistent connection and pipelined requests.

Figure (a) shows non-persistent connections. Browser makes connection and request for each object. Server parses request, responds and closes connection. It has performance problem.

Figure (b) shows persistent connections. On same TCP connection server parses request, responds, parses new request upon responds and so on . It improves performance.

Figure (c) shows pipelining with persistent connections. It send next request before previous response is received. Pipelining with persistent connection improves performance

HTTP Example



FTP (File Transfer Protocol)

It allows a user to copy files to/from remote hosts. FTP uses the services of TCP. It needs two TCP connections. The well-known port 21 is used for the control connection and the well-known port 20 for the data connection

Goal of FTP (Objective of FTP)

- promote sharing of files
- encourage indirect use of remote computers
- shield user from variations in file storage
- transfer data reliably and efficiently

Working of FTP

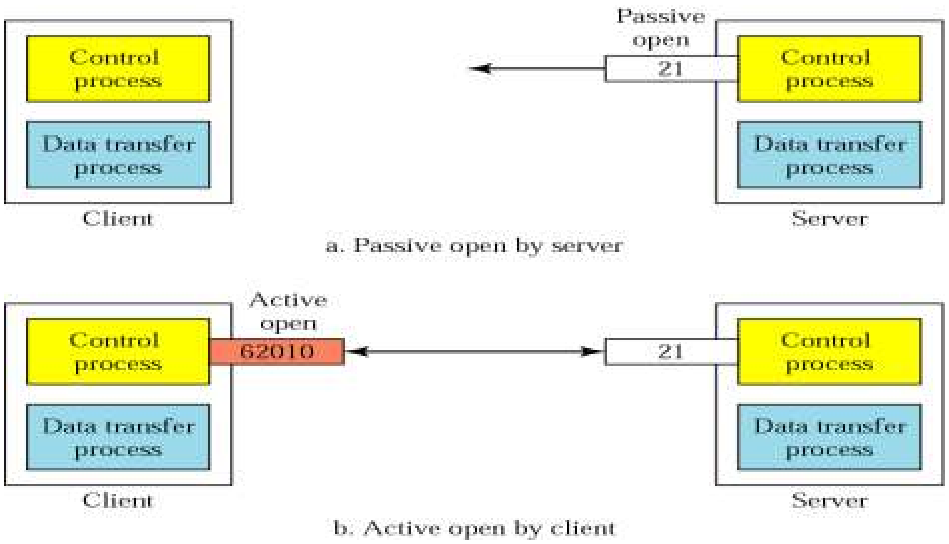


Figure: FTP Connections- control connection

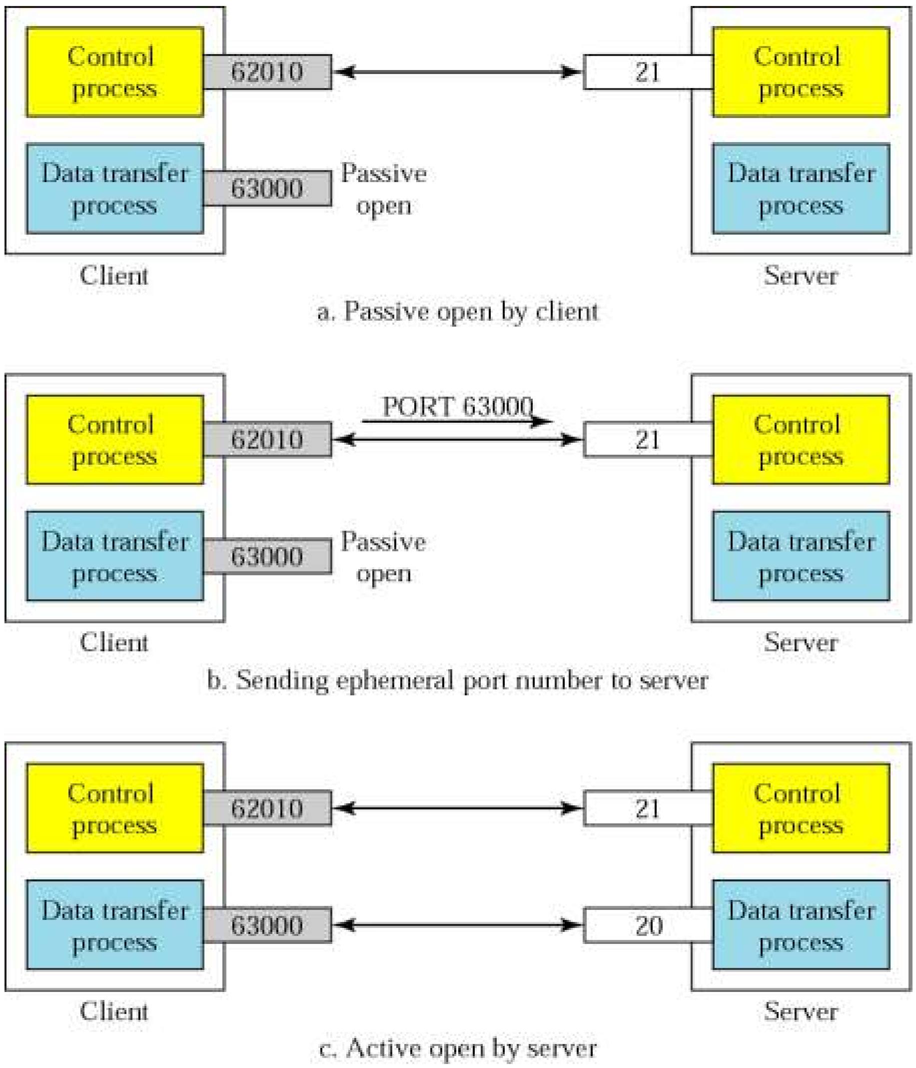
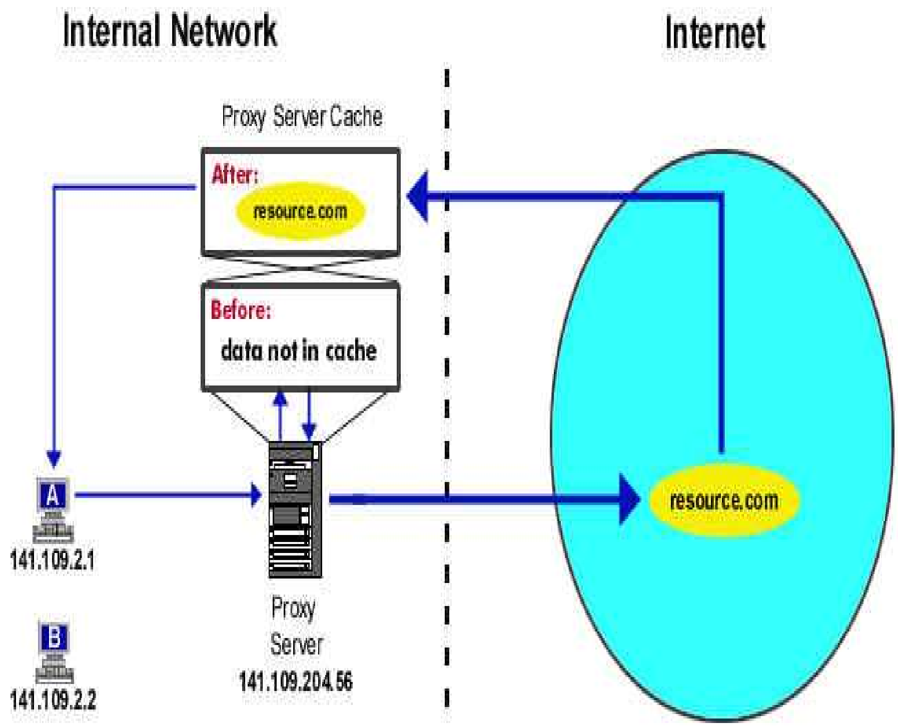


Figure: FTP connection- data connection

- Uses Server's well-known port 20
- Client issues a passive open on an ephemeral port, say *x*.
- Client uses PORT command to tell the server about the port number *x*.
- Server issues an active open from port 20 to port *x*.
- Server creates a child server/ephemeral port number to serve the client

Proxy Server

Proxy server is a server (a computer system or an application) that acts as an intermediary for requests from clients seeking resources from other servers.Proxy server stores all the data it receives as a result of placing requests for information on the internet in it's *cache.*Cache simply means memory. The cache is typically hard disk space, but it could be RAM. Caching documents means keeping a copy of internet documents so the server doesn't need to request them over again. With proxy caching, clients make requests to servers, but the requests first go through a proxy cache.

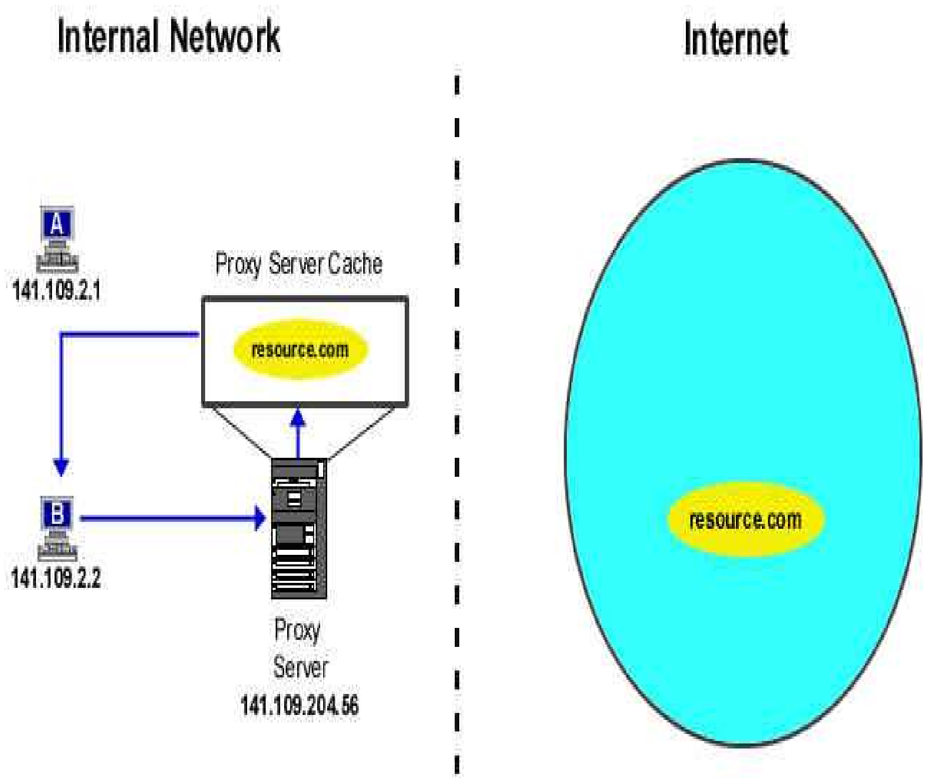


Scenario 1: Caching a Document on a Proxy Server

User **A** request a web page. The request goes to the proxy server, the proxy server checks to see if the document is stored in cache. If the document is not in cache so the request is sent to the Internet. The proxy server receives the request, stores (or caches) the page. The page is sent to user **A** where is viewed.

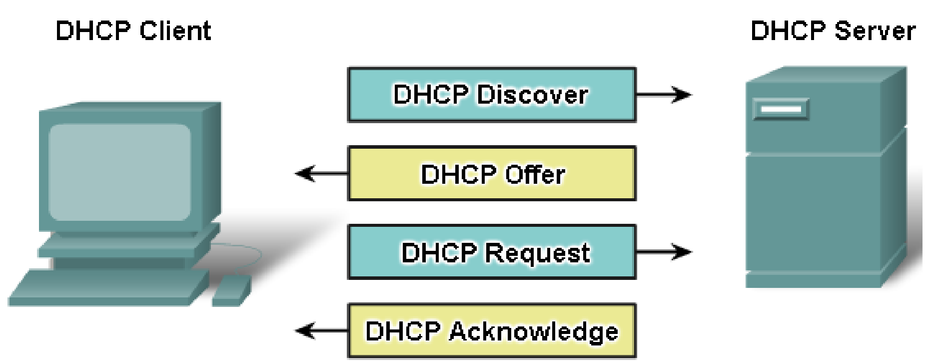
Scenario 2: Retrieving Cached Documents

User **B** request the same page as user **A** (ie. resource.com).The request goes to the proxy server. The proxy server checks its cache for the page. Since the page is stored in cache, the proxy server sends the page to user **B** where it is viewed. Connection to the Internet is required



DHCP (Dynamic Host Configuration Protocol)

When a DHCP-configured device boots up or connects to the network, the client broadcasts a DHCP DISCOVER packet to identify any available DHCP servers on the network. A DHCP server replies with a DHCP OFFER, which is a lease offer message with an assigned IP address, subnet mask, DNS server … etc. The client might choose from multiple DHCP OFFERs it receives and broadcast a DHCP REQUEST. If the DHCP OFFER is still available the DHCP server will reply with DHCP ACKNOWLEDGE. If the DHCP OFFER is not available the server will reply with DHCP NAK the process will start all over again



Electronic Mail

It is one of the most widely used and popular internet applications. User can communicate with each other across the network. Every user owns his own mailbox which he uses to send, receive and store messages from other users. Every user can be uniquely identified by his unique email address. Mailbox principle-A sender does not require the receiver to be online nor the recipients to be present. A user’s mailbox can be maintained anywhere in the internet on the server.

Three major components: user agents, mail servers and Simple Mail Transfer Protocol (SMTP)

<u>User Agent</u>:“mail reader”

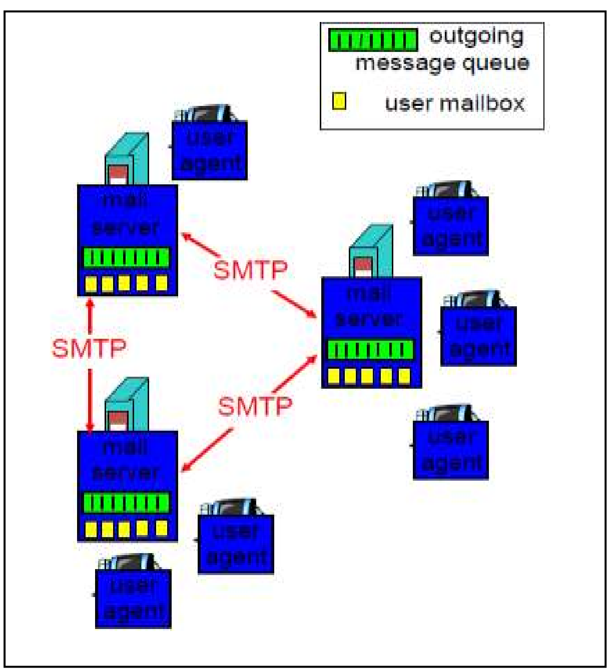
It consist of composing, editing, reading mail messages. e.g., Eudora, Outlook, elm, Netscape Messenger. All the outgoing, incoming messages are stored on server

<u>Mail Servers</u>

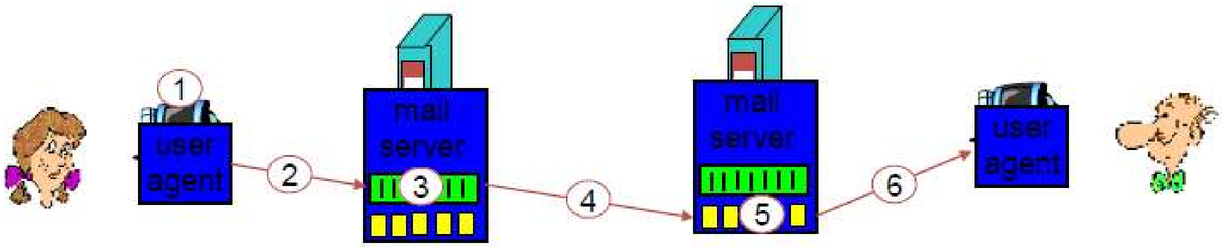
Mailbox contains incoming messages for user. The message queue of outgoing (to be sent) mail messages

<u>SMTP</u>

It is protocol between mail servers to send email messages; client: sending mail server and “server”: receiving mail server. It uses TCP to reliably transfer email message from client to server. It uses port 25.There are three phases of transfer: handshaking (greeting), transfer of messages and closure.



Example



Alice uses UA to compose message and send "to" bob@someschool.edu . Alice's UA sends message to her mail server. The message is placed in message queue. Client side of SMTP opens TCP connection with Bob's mail server. SMTP client sends Alice's message over the TCP connection. Bob's mail server places the message in Bob's mailbox. Bob invokes his user agent to read message.

POP (Post Office Protocol)

It is a protocol used to retrieve e-mail from a mail server. Most e-mail applications (sometimes called an *e-mail client*) use the POP protocol. Examples of native mail system are MS outlook, Lotus Notes, MS Exchange, Eudora. There are two versions of POP: The first, called *POP2,* became a standard in the mid-80's and requires SMTP to send messages. The newer version, POP3, can be used with or without SMTP. POP3 uses TCP/IP port 110.

IMAP (Internet Message Access Protocol)

It is a method of accessing electronic mail messages that are kept on a possibly shared mail server. In other words, it permits a "client" email program to access remote message stores as if they were local. For example, email stored on an IMAP server can be manipulated from a desktop computer at home, a workstation at the office and a notebook computer while travelling. IMAP uses TCP/IP port 143.

POP vs IMAP

Feature	POP3	IMAP
Where is protocol defined?	RFC 1939	RFC 2060
Which TCP port is used?	110	143
Where is e-mail stored?	User's PC	Server
Where is e-mail read?	Off-line	On-line
Connect time required?	Little	Much
Use of server resources?	Minimal	Extensive
Multiple mailboxes?	No	Yes
Who backs up mailboxes?	User	ISP
Good for mobile users?	No	Yes
User control over downloading?	Little	Great
Partial message downloads?	No	Yes
Are disk quotas a problem?	No	Could be in time
Simple to implement?	Yes	No
Widespread support?	Yes	Growing